Project Initialization and Planning Phase

Date	29 June 2026
Team ID	SWTID-2026-8096
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project aims to develop an intelligent flood prediction system using Machine Learning techniques to provide accurate and timely flood forecasts. The system analyzes historical weather and rainfall data to identify flood-prone conditions, enabling disaster management authorities to take preventive measures. By improving prediction accuracy and reducing response time, the project helps minimize the loss of lives and property during flood events. The solution is deployed through a Flask web application and is designed for cloud deployment using IBM Cloud.

Project Overview	
Objective	The primary objective of this project is to develop a Machine Learning based Flood Prediction System that accurately predicts flood occurrences using meteorological parameters such as annual rainfall, seasonal rainfall, cloud cover, and visibility. The system aims to provide early warnings that support effective disaster preparedness and emergency response.
Scope	The project focuses on collecting and preprocessing historical weather datasets, training multiple Machine Learning classification models, evaluating their performance, and deploying the best-performing model through a Flask web application. The application can be hosted on IBM Cloud to provide real-time flood prediction services for disaster management authorities.
Problem Statement	

Description	Floods are among the most devastating natural disasters, causing significant damage to infrastructure, agriculture, and human lives. Traditional flood forecasting methods often fail to provide timely and accurate predictions due to limited data analysis capabilities. There is a need for an intelligent prediction system capable of analyzing multiple weather parameters to generate reliable flood warnings.
Impact	An accurate flood prediction system enables early evacuation planning, efficient allocation of emergency resources, reduced economic losses, and improved public safety. It assists government agencies and disaster management organizations in making informed decisions during flood emergencies.
Proposed Solution	
Approach	The proposed system uses Machine Learning to analyze historical weather data and predict the possibility of floods. It trains multiple models such as Decision Tree, Random Forest, KNN, and XGBoost, and selects the best performing model. The selected model is integrated into a Flask web application to provide quick and accurate flood predictions.
Key Features	• Machine Learning-based flood prediction system. • Analysis of rainfall, seasonal rainfall, cloud cover, and visibility. • Comparison of multiple classification algorithms. • XGBoost model with **96.55% prediction accuracy**. • Real-time prediction using a Flask web application. • Cloud deployment support using IBM Cloud. • User-friendly interface for disaster management authorities.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 Processor or equivalent
Memory	RAM specifications	8 GB

Storage	Disk space fordata,models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm, Visual Studio Code
Data		
Data	Source, size, format	Historical Flood Prediction Dataset, Kaggle, CSV, Attributes: Rainfall, Seasonal Rainfall, Cloud Cover, Visibility, Flood Status
Input Attributes	Features used for prediction	Annual Rainfall, Jan–Feb Rainfall, Mar–May Rainfall, Jun–Sep Rainfall, Oct–Dec Rainfall